AI Based Smart Retail Recommendation System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The retail sector is evolving from static, transaction-based recommendation systems to dynamic and context-aware systems that aim to understand customer behavior and emotional states. Traditional models mainly rely on past purchase data and often ignore real-time customer conditions, which limits their ability to deliver truly personalized shopping experiences. This research identifies an important gap in existing studies. Many previous works depend on camera-based facial expression analysis to detect emotions, which raises privacy concerns and is not always suitable for real-world deployment. In addition, these systems often fail to combine emotional insights with detailed behavioral analysis and real-time decision-making mechanisms.

To address this issue, this research proposes a privacy-preserving multimodal framework that combines non-visual indoor positioning and mobile-based emotional input to identify customer states. The system uses Bluetooth Low Energy (BLE) technology to track customer movement patterns such as walking paths, navigation behavior, and time spent in different store sections. At the same time, customer emotional states are captured through a mobile application or connected components, such as user feedback or wearable devices, instead of using cameras. This combination of behavioral and emotional data provides a more complete understanding of shopper intent.

By integrating these technologies into a single intelligent decision-support system, retailers can move from reactive marketing approaches to proactive customer assistance. The system can detect early signs of confusion, hesitation, or frustration and automatically notify store staff to provide timely support. This approach can improve customer satisfaction, increase purchase conversions, and enhance overall shopping efficiency.

Furthermore, the proposed system supports a privacy-friendly and human-centered retail environment, as it avoids the use of CCTV and facial recognition technologies. By combining digital intelligence with user-controlled emotional input, the system ensures ethical data usage while still delivering personalized in-store experiences.

Keywords: Behavioral Analytics, Indoor Positioning Systems, BLE Beacons, Retail Recommendation Systems, Multimodal Data Fusion, Privacy-Preserving Systems

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1. INTRODUCTION

1.1 Background and Context

The retail industry is changing quickly due to new technologies and changing customer expectations. Today, retailers use technologies like artificial intelligence (AI) and recommendation systems to improve customer experience and increase sales [1], [5]. In Sri Lanka, the retail sector is also growing with the development of shopping malls, supermarkets, and online platforms, creating new opportunities for digital transformation [2], [3]. However, many existing retail systems still depend on past purchase data and fixed customer profiles. Because of this, they cannot understand what customers are doing or feeling in real time while shopping.

Understanding customer behavior and emotions is important because they affect buying decisions and overall satisfaction. For example, customers may feel confused, frustrated, or interested while shopping, and these feelings influence their actions [4]. In the past, researchers used visual methods like facial expressions and body language to understand these emotions. However, these methods usually require cameras, which can create privacy concerns and may not be suitable for real retail environments.

To solve this problem, researchers have started using non-visual technologies to track customer behavior. One popular method is the use of indoor positioning systems (IPS). Technologies like Bluetooth Low Energy (BLE) beacons can track customer location, movement, and time spent in different areas without using cameras [6], [8], [10]. These systems work by measuring signal strength (RSSI) from beacons placed around the store to identify where the customer is [7].

Researchers have also improved the accuracy of BLE systems by using methods such as signal filtering and combining BLE with other techniques like Pedestrian Dead Reckoning (PDR) [13], [18], [19]. These systems help retailers understand how customers move inside the store and which areas are most popular. However, most of these systems only track movement and do not capture customer emotions.

Therefore, there is a need for a system that can combine customer behavior with emotion detection without using cameras. This research proposes a system that uses BLE-based tracking together with mobile-based emotional input and behavior analysis. This approach can help retailers understand customers better in real time while also protecting their privacy and improving the shopping experience.

1.2 Literature Review

Many researchers have studied how to improve retail systems by understanding customer behavior and decision-making. Traditional retail systems mainly use past purchase data and customer profiles to make recommendations. However, these methods cannot understand what customers are doing in real time inside the store. Newer studies focus on analyzing the customer journey, including how customers move, interact with products, and make decisions while shopping [4], [5].

Customer behavior can be understood using factors such as time spent in a section (dwell time), movement between store areas, and navigation paths. Research shows that these factors can help identify customer interests and decision patterns. For example, if a customer stays in one area for a long time or moves back and forth between sections, it may indicate confusion or difficulty in choosing products [22], [23]. These insights are useful for improving store layout and customer experience.

To track customer movement inside stores, many researchers use Indoor Positioning Systems (IPS). These systems use technologies such as Bluetooth Low Energy (BLE), Wi-Fi, and hybrid methods to locate customers in indoor environments where GPS cannot work properly [6], [21]. Among these, BLE beacon-based systems are widely used because they are low-cost, energy-efficient, and easy to install [8], [10]. These systems use signal strength values (RSSI) to estimate the customer's location and identify the store zone they are in [7].

Several studies have improved BLE-based systems by using techniques such as signal filtering, smoothing, and machine learning to increase accuracy [7], [18]. Some research also combines BLE with other methods like Pedestrian Dead Reckoning (PDR) to get better location results [13], [17]. These systems are used in retail environments to track customer movement, analyze shopping patterns, and generate insights such as store traffic and popular areas [20].

However, most existing systems mainly focus on tracking movement and analyzing data for business insights. They do not identify whether a customer is facing difficulty while shopping. Also, many systems do not include emotional understanding or real-time assistance. Camera-based emotion detection methods exist, but they raise privacy concerns and are not always practical in real stores. Therefore, there is a need for a system that combines behavior tracking with alternative emotion detection methods, such as mobile-based input, and provides real-time support to customers. This research aims to address this gap by integrating BLE-based tracking, behavioral analysis, and mobile-based emotion input into a single system.

1.3 **Research Gap**

Many studies have used artificial intelligence (AI) and machine learning to understand customer behavior in retail stores. These systems mainly focus on tracking customer movement, navigation paths, and interaction patterns to understand how customers shop and interact with products [1], [5]. Researchers have also used behavior analysis methods such as dwell time, heatmaps, and indoor positioning systems to study customer shopping patterns and preferences [22], [23]. In addition, technologies such as Bluetooth Low Energy (BLE) and Wi-Fi are used to track customer location and movement inside stores without using cameras [6], [8]. These approaches help retailers improve store layout and marketing strategies.

However, most existing systems focus mainly on analyzing behavioral data for business insights and do not provide real-time support to customers. Although some studies have explored customer emotions, many of them rely on camera-based methods such as facial expression detection, which raises privacy concerns and is not always suitable in real retail environments [4]. At the same time, non-visual tracking systems do not capture emotional states, and they often fail to combine behavior and emotion data to understand customer difficulties. As a result, these systems cannot effectively identify situations such as confusion, hesitation, or frustration during shopping.

Therefore, there is a clear research gap in developing a system that combines non-visual customer tracking with alternative emotion detection methods and real-time assistance. This research proposes a solution that integrates BLE-based indoor positioning with mobile-based emotional input and behavioral analysis to detect customer difficulties. The system also provides real-time notifications to nearby store staff, allowing them to assist customers at the right time. This approach improves the overall shopping experience while maintaining customer privacy and supporting more intelligent retail decision-making [3].

Table 1: Current research comparison 1

Feature	Traditional Retail Systems [1][2]	Retail Analytics & Recommendation Systems [3][5]	Indoor Positioning Systems (BLE-Based) [6][8][10]	Behavior Tracking & Movement Analysis [22][23]	Proposed System
Detect facial emotions (CCTV)	✕	✕	✕	✕	✕
Track customer movement	✕	✓	✓	✓	✓
Analyze shopping behavior	✕	✓	✓	✓	✓
Provide retail recommendations	✓	✓	✕	✕	✓
Real-time assistance to customers	✕	✕	✕	✕	✓
Notify store staff automatically	✕	✕	✕	✕	✓
Capture emotions (non-visual / mobile-based)	✕	✕	✕	✕	✓
Use non-visual tracking (BLE/Wi-Fi)	✕	✕	✓	✓	✓
Identify customer difficulty (confusion/hesitation)	✕	✕	✕	limited	✓

Combine emotion + behavior analysis	✕	✕	✕	✕	✓
Real-time behavior analysis	✕	limited	✓	✓	✓

2. OBJECTIVE

2.1 Main Objective

This project aims to develop an AI-powered system that uses non-visual indoor positioning, customer behavior analysis, and mobile-based emotion input to track customer movement and shopping patterns in real time, identify customers who experience confusion, hesitation, or frustration, and automatically notify store staff to provide timely assistance and improve the overall retail shopping experience.

2.2 Specific Objective

1. To develop a non-visual indoor positioning system using technologies such as Bluetooth Low Energy (BLE) to track customer location inside the retail store.
2. To analyze customer behavior by monitoring movement patterns, navigation paths, and time spent in different store sections.
3. To capture customer emotions through a mobile application or connected interface (such as user input or feedback).
4. To identify customers who face shopping difficulties such as hesitation, confusion, repeated movement, or long dwell time by combining behavior and emotion data.
5. To integrate the behavior and emotion detection module with the smart retail recommendation system to improve personalized product suggestions.
6. To design and implement a real-time automated notification system that alerts nearby store staff when customers need assistance.
7. To evaluate the system performance by measuring the accuracy, response time, and effectiveness of behavior and emotion detection in real-time retail environments.

3. METHODOLODGY

The proposed system follows a Design Science research approach, where the main goal is to design and develop a practical AI-based solution for detecting customer emotions and shopping behavior in retail stores. Instead of using camera-based methods, the system uses non-visual indoor positioning and mobile-based emotion input to understand customer conditions in real time. Bluetooth Low Energy (BLE) beacons installed in different areas of the store will be used to track customer location. A mobile application or smart device will collect signal strength (RSSI) data from nearby beacons to determine the customer's position within store zones.

In addition, customer behavior will be analyzed by tracking movement patterns, navigation paths, and the time customers spend in different sections of the store. Behavioral indicators such as dwell time, repeated movement between areas, and navigation patterns will be used to understand customer actions. At the same time, customer emotions will be captured through a mobile application interface, where users can provide simple feedback or emotional inputs, or through connected components such as wearable devices. These emotion and behavior data will be combined to identify customers who may face difficulties such as confusion, hesitation, or frustration during shopping.

The collected data will be processed using rule-based logic and machine learning techniques. The system will analyze both behavioral patterns and emotional inputs to calculate a difficulty score for each customer. When the system detects a high level of difficulty, it will automatically send a notification to nearby store staff through a mobile application, including the customer's location and type of issue. This enables staff to provide timely assistance and improve the shopping experience.

The system will be developed using technologies such as Python for machine learning, mobile application frameworks (Flutter or Android) for data collection, and backend platforms such as Node.js or Firebase for real-time data processing. Machine learning libraries such as Scikit-learn or TensorFlow will be used to develop behavior classification models. Simulated movement data and user input data will be used for testing and validation. During data collection, ethical guidelines will be followed to protect customer privacy, and personal identity information will not be stored.

The system performance will be evaluated through experimental testing using common performance metrics.

Validation metrics include:

- **Accuracy** – to measure how correctly the system detects customer behavior and emotions
- **Precision and Recall** – to evaluate the reliability of detection results
- **F1-score** – to measure the overall model performance

• **Response Time** – to check how quickly the system can detect customer difficulties and notify store staff

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and its Components

The proposed system is a Smart Retail Recommendation System that improves the customer shopping experience using artificial intelligence and data analysis. The system collects and analyzes customer information in real time and provides personalized recommendations and assistance inside the retail store.

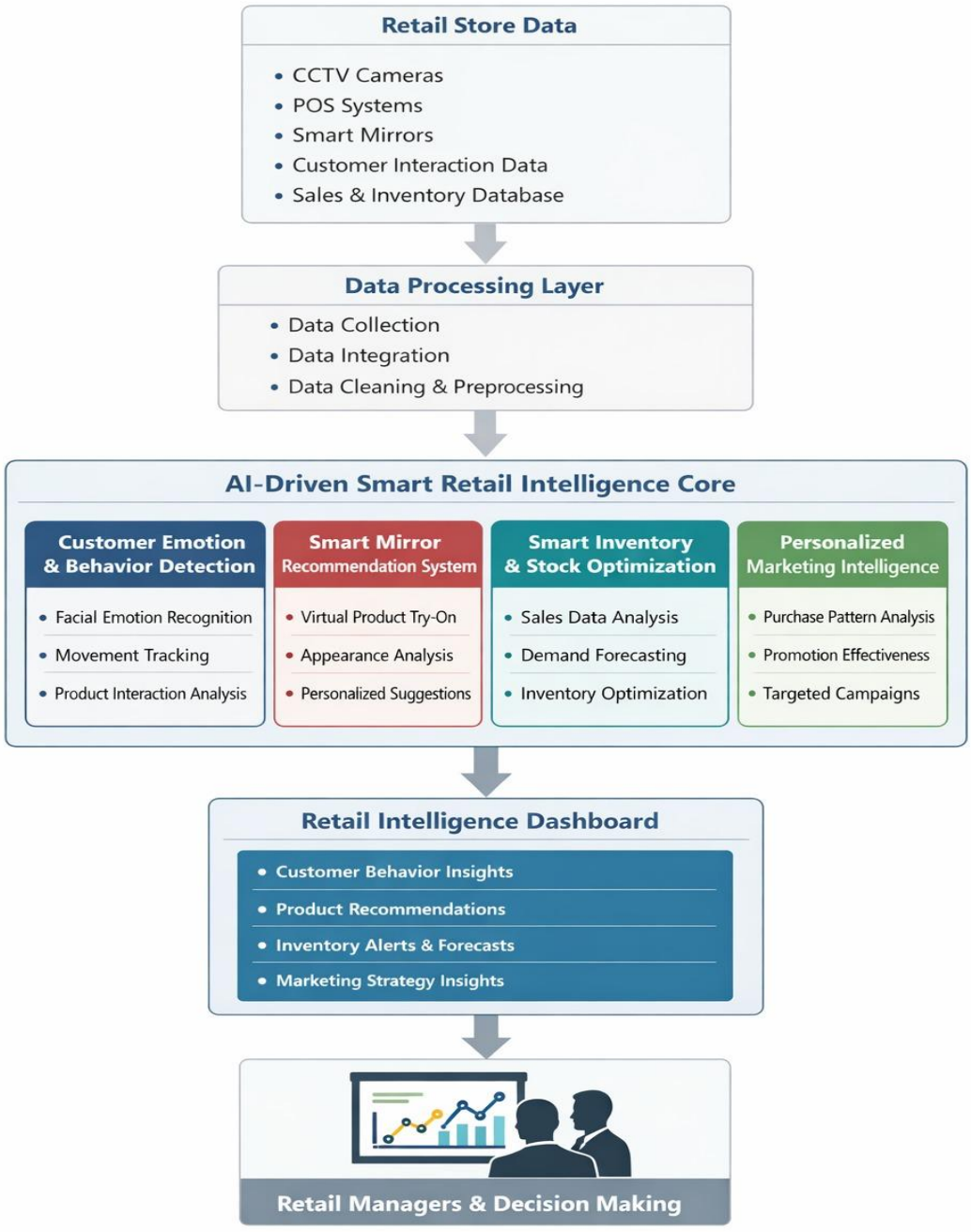
The overall system contains four main components:

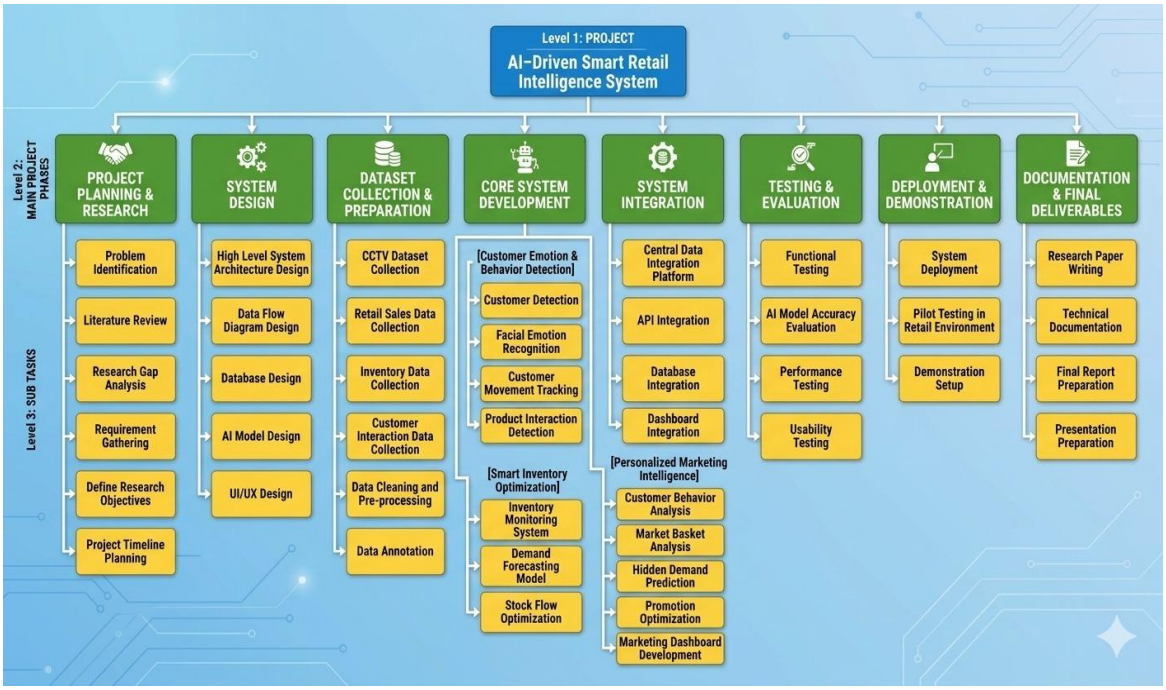
1. Customer Emotions and Behavior Detection – detects customer emotions and shopping behavior using computer vision and AI.
2. Smart Mirror Recommendation System – provides product recommendations and virtual try-on features using a smart mirror interface.
3. Smart Inventory and Stock Flow Optimization – monitors product stock levels and manages inventory efficiently.
4. Personalized Marketing Intelligence analyzes customer data and generates targeted promotions and marketing insights.

These components work together to create a smart and personalized retail environment.

Figure 1: System Overview Diagram

Figure 2: Data Flow Diagram





4.2 **Component Interaction**

All four components in the system interact with each other.

- The Emotion and Behavior Detection component identifies customer emotions and shopping patterns.
- This information can be shared with the Smart Mirror Recommendation System to provide better product suggestions.
- The Inventory System checks product availability when recommendations are generated.
- The Marketing Intelligence component uses customer behavior data to create personalized promotions.

This interaction allows the system to deliver better recommendations and improved customer service.

4.3 Scalability and Security Consideration

The system is designed to support multiple customers at the same time in a retail environment. Cloud services and scalable AI models can be used to handle large volumes of image and behavior data.

To protect customer privacy, the system follows data protection and security guidelines. Personal identity information will not be stored, and image data will be processed securely. Access control, secure storage, and encryption methods will be used to ensure that customer data remains safe.

5. User Requirements

5.1 Stakeholder Identification

The system will be used or affected by several stakeholders in the retail environment.

- Customers – People who visit the retail store and interact with products and services.
- Retail Store Staff – Employees who assist customers and receive alerts when customers need help.
- Store Managers – People who monitor store performance and customer behavior insights.
- System Administrators – Technical staff responsible for maintaining the system and ensuring it works properly.

5.2 Functional Requirements

The system should provide the following main functions:

• The system shall track customer location inside the store using non-visual indoor positioning technologies such as Bluetooth Low Energy (BLE).

• The system shall collect signal data (RSSI) through a mobile application or connected device to identify the customer's current store zone.

• The system shall capture customer emotional input through a mobile application or connected interface (such as user feedback or simple emotion selection).

• The system shall analyze customer movement and behavior**,** including navigation paths, time spent in different sections (dwell time), and movement patterns.

• The system shall combine behavior data and emotion input **to** detect customer states such as confusion, hesitation, or frustration.

• The system shall identify customers who may need assistance based on detected behavior patterns and emotional signals.

• The system shall send real-time notifications to nearby store staff**,** including customer location and type of issue.

• The system shall store customer behavior and interaction data for further analysis and improvement of retail strategies.

- The system shall integrate with other smart retail components**,** such as recommendation systems, to improve personalized services

5.3 **Non-Functional Requirements**

Security: The system must protect customer privacy and store data securely. Only authorized users should access the system.

Performance: The system should detect customer emotions and behaviour in real time and handle multiple customers at the same time.

Usability: The system interface should be simple and easy for store staff and managers to use. Notifications should be clear.

Reliability: The system should run continuously during store operating hours without failures.

Scalability: The system should support expansion to larger stores or multiple retail locations in the future.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The target market for the Smart Retail Recommendation System includes retail businesses such as supermarkets, shopping malls, clothing stores, and large retail chains. The main users will be storing owners, store managers, and retail technology providers who want to improve customer experience, increase sales, and better understand customer behavior.

6.2 Value Proposition

The system provides a smart and personalized shopping experience by combining technologies such as emotion detection, smart mirror recommendations, inventory management, and personalized marketing. This helps retailers understand customer needs in real time and improve product recommendations and sales.

6.3 Revenue Model

Revenue can be generated through:

- Software subscription fees (monthly or yearly)
- System installation and setup services
- Maintenance and technical support services
- Retail data analytics services

6.4 Cost Estimation

Main costs include:

- Software development for AI models and system platform
- Hardware costs such as cameras and smart mirrors
- Cloud storage and computing
- System maintenance and support

- **Pricing Strategy**

The system will use a tier-based pricing model:

- Basic Package – Basic recommendation features for small stores
- Standard Package – Includes emotion detection and smart mirror features
- Premium Package – Full system with advanced analytics and real-time assistance

- **Competitive Advantage**

The system offers:

- Integration of multiple smart retail technologies
- Real-time emotion and behaviour detection
- Personalized product recommendations
- Staff notifications for customer assistance

- **Intellectual Property Considerations**

The system can be protected through:

- Software copyright for the developed system
- Patents for unique technologies
- Trademark registration for the product name and brand.

7. BUDGET AND JUSTIFICATION

7.1 Hardware costs

Table 2:Hardware Costs

Item	Estimated Costs
Laptop / Desktop PC	LKR 215,000

***Justification*:** Hardware resources are required to develop, train, and test the machine learning models.

7.2 Software Licenses

Most tools used in this research are open-source.

EX: Python

TensorFlow / PyTorch

Scitkit-Learn

PostgreSQL / MongoDB

7.3 Cloud Services

Table 3:Cloud Services Costs

Service	Cost
Cloud Computing for AI training	LKR 35000.00
Cloud storage	LKR 25000.00

***Justification*:** Cloud services enable scalable data storage and model deployment.

7.4 Human Effort Cost

- Since this is an academic research project development is carried out by the student research team.
- Estimated academic effort value.

Table 4: Human Effort Cost

Role	Estimated Value
Research & Development	LKR 20000.00

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) is used to identify tasks involved in developing the Smart Inventory and Stock Flow Optimization module of the proposed system. This module is designed to improve the accuracy of demand forecasting, optimize stock movement between stores and warehouses, and predict transportation costs using artificial intelligence techniques

Table 5: Work Breakdown Structure

Task ID	Task Name	Description
T1	Literature Review	Study existing research on demand forecasting, inventory optimization, and explainable AI
T2	Dataset Collection	Gather simulated retail sales datasets and contextual data such as weather and seasonal trends
T3	Data Preprocessing	Clean datasets, remove inconsistencies, and prepare structured data for model training
T4	Feature Engineering	Create features such as seasonal demand indicators, weather influence, and product sales patterns
T5	Demand Forecasting Model Development	Develop machine learning model to predict product demand
T6	Model Training & Evaluation	Train models and evaluate prediction accuracy

T7	Stock Flow Optimization Design	Design algorithm to identify stock shortages and excess stock
T8	Stock Transfer Recommendation System	Develop logic to recommend stock transfer between stores and warehouses
T9	Transport Cost Prediction Model	Develop AI model to estimate logistics cost for stock Transfers
T10	Explainable AI Integration	Implement explainable AI methods to provide reasoning behind predictions
T11	System Integration	Integrate forecasting, optimization and cost prediction modules
T12	Testing and Validation	Test System performance using simulated datasets
T13	Retail Analysis	Analyze prediction accuracy and optimization performance
T14	Documentation	Prepare research documentation and evaluation reports

8.1 Timeline Alignment

The development of the Smart Inventory and Stock Flow Optimization module will be conducted over a period of six months

Phase	**Activity**
Phase 1	Literature review and research planning
Phase 2	Dataset collection and preprocessing
Phase 3	Demand forecasting model development

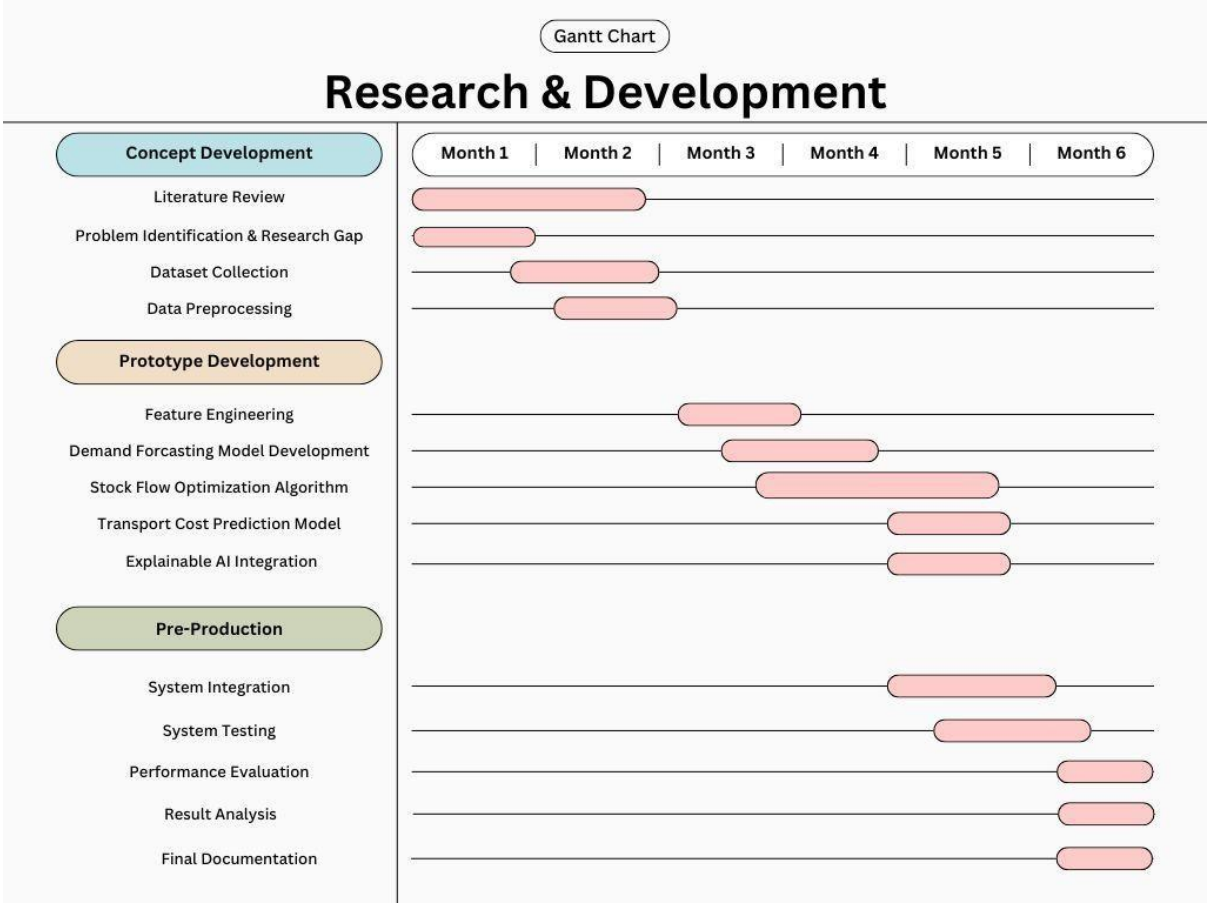
Phase 4	Stock flow optimization and transport cost prediction
Phase 5	System testing and evaluation
Phase 6	Documentation and final research report

8.2 Balanced workload Distribution

The activities involved in the Smart Inventory and Stock Flow Optimization module are spread across different research phases, ensuring a balanced workload for the entire project timeline. The development process for this module consists of data preparation, developing the machine learning model, optimization algorithm implementation, and system evaluation.

9. GANTT CHART

Figure 4: Gantt Chart



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11. APPENDICES

11.1 Appendix A - Survey Questions

1. What is your age group?
2. How often do you visit retail clothing stores?
3. Do you usually spend a long time deciding what to buy?
4. Have you ever felt confused while choosing products in a store?
5. Do you move between different sections multiple times before making a decision?
6. How often do you need help from store staff while shopping?
7. What emotions do you usually feel while shopping?
8. Would you be willing to select your emotion in a mobile app while shopping?
9. Do you think your emotions affect your buying decisions?
10. Are you comfortable with a system that tracks your movement using Bluetooth (no camera)?

11.2 Appendix B – Ethical Considerations

This research follows ethical guidelines to ensure responsible data usage and system development.

• The research uses simulated retail data rather than confidential company data.

• No personal customer information will be collected or analyzed.

• The system focuses only on product inventory and sales patterns.

• Any survey responses collected will remain anonymous and used solely for academic purposes.

• The research does not involve sensitive personal data or harmful technologies.

These measures ensure that the research maintains data privacy, confidentiality, and ethical

11.3 CONCLUSION

In conclusion, this proposal presents the development of a Customer Emotions and Behavior Detection component, which is an important part of the Smart Retail Recommendation System. The main goal of this component is to improve the in-store shopping experience by detecting customer emotions and behavior in real time using artificial intelligence and computer vision technologies.

The proposed system analyzes customer facial expressions, movements, and shopping behavior inside the retail store. By understanding emotions such as confusion, hesitation, or interest, the system can identify customers who may need assistance during their shopping journey. When such situations are detected, the system can notify store staff so they can provide timely support and improve customer service.

This component represents an important step toward creating a more intelligent and customer-focused retail environment. Instead of only collecting customer data for analysis, the system actively responds to customer needs and improves the shopping experience. Overall, this solution can help retailers better understand customer behavior, increase customer satisfaction, and support the future development of AI-powered smart retail systems.